

COMP3052.SEC Coursework 2 (20%)

Portfolio Part2 (based on lab sessions)

Learning Objectives

This coursework is designed to enable you to reflect and report on your computer security experiences in the lab sessions. You should also use appropriate references.

Explanation

You are required to write a second detailed report, of about 1000 words, that covers a second aspect of computer security that you will have encountered in the labs and lectures. Marks will be awarded based on:

- Correctness and completeness of your answers
- Depth of exploration and technical accuracy
- Quality of justification and evidence
- Integration of personal reflection and practical experience

Topic: Firewalls and SSH Tunneling

You are expected to write **up to 1000 words (± 100)** for the following topic (excluding references):

It has become commonplace to use permitted services such as SSH to “tunnel” traffic that would otherwise be blocked by a network firewall.

- Describe in detail **why a network administrator might choose to block certain ports** and how this forms part of broader security policy.
- Explain **how a protocol such as SSH can be used to circumvent firewall restrictions**, including the mechanics of tunneling.
- Provide **one example of a legitimate use case** for SSH tunneling.
- Provide **one example of a disreputable or malicious use**.
- Discuss the **security implications** of allowing or blocking tunneling, and how administrators might detect or mitigate misuse.
- Evaluate the **technical methods administrators can use to detect SSH tunneling activity**. What are the strengths and limitations of these approaches?

To achieve top marks, it is essential to substantiate your arguments with evidence from your own practical lab experiments. Include screenshots of these experiments in the appendix of your report to demonstrate your hands-on engagement and understanding. Please note that screenshots taken directly from the provided lab materials will not be considered as your own work and will not contribute to your assessment. Any additional knowledge beyond what you have learned in the class sessions or labs will demonstrate that you have really understood the concepts.

Additionally, when proposing your SSH tunnelling method, ensure that your recommendations are well-supported by referencing credible sources. Cite articles from reputable journals or academic publications to strengthen the validity and reliability of your method. This will demonstrate your ability to integrate theoretical knowledge with practical insights, showcasing a comprehensive understanding of the topic.

Submission & Deadline

The deadline for submission is 17:59:59, May 11. All submissions must be through Moodle.